PTS

Pan / Tilt / Slider

A five-mount motorised camera system for live video production - wireless, self-healing, and controllable from everything in the booth.

PTS drives five motorised camera mounts over a dedicated wireless network. Each mount pairs a Teensy 4.1 motion controller (TMC2209 silent stepper drivers, StallGuard homing) with an ESP32 bridge and round touchscreen; a central hub links every mount to the operator's control surfaces. Positions, speed presets and subject calibrations live on the mounts themselves - controllers can come and go mid-show.

System architecture

Control surfaces
PC app (PyQt6, joystick, CV tracking) · web app on phones · 7" hub touchscreen · Bitfocus Companion + QLab via OSC

- - - - LAN / WiFi / USB - - - -

HUB - ESP32-S3
WiFi AP 'CamMount' · TCP / WebSocket / OSC servers · USB serial to PC · UART to hub display · autonomous self-recovery

- - - - ESP-NOW wireless star (touch-screen pairing, channel self-healing) - - - -

CAM 1	CAM 2	CAM 3	CAM 4	CAM 5
ESP32 AMOLED bridge	ESP32 AMOLED bridge	ESP32 AMOLED bridge	ESP32 AMOLED bridge	ESP32 AMOLED bridge
Teensy 4.1 + TMC2209	Teensy 4.1 + TMC2209	Teensy 4.1 + TMC2209	Teensy 4.1 + TMC2209	Teensy 4.1 + TMC2209
pan · tilt · slider · zoom	pan · tilt · slider · zoom	pan · tilt · slider · zoom	pan · tilt · slider · zoom	pan · tilt · slider · zoom

Highlights

Feature	What it does
Look-at tracking	Camera keeps aiming at a calibrated 3D subject while the slider travels - subjects are calibrated in two taps and switchable mid-move from any controller.
Zero-config pairing	No MAC addresses or per-unit firmware: hold a mount's screen, pick CAM 1-5, tap your hub. The hub learns mounts on first contact; conflicts resolve with one tap on the hub display.
Self-healing	Autonomous recovery ladders on hub and mounts (reinit then restart), idle maintenance restarts, watchdogs at every layer, and a motion-side dead-man: a mount stops within 500 ms of losing its control stream.
Health telemetry	Every node reports heap, loop timing, link quality and reset cause every 10 s into the PC log - degradation is visible hours before it becomes a symptom.
Show-control ready	OSC servers on both the hub and the PC app put every function on Stream Deck buttons and QLab cue stacks (reference overleaf).

OSC control reference - Companion / QLab

Two identical OSC servers listen on **UDP port 9700** and accept the same address space, so Companion pages work unchanged against either target:

Target	Address	When to use
The hub itself	`192.168.4.1`	No PC needed - hub + display + mounts + Stream Deck is a complete rig. Reach it via a WiFi-to-LAN bridge joined to the CamMount AP, or any machine joined to CamMount directly.
PC app machine	`that PC's IP`	When the PC app is running anyway. Configurable via osc_enabled / osc_port in the app config.

Companion: add a **Generic: OSC** connection with the target IP and port 9700, then use its Send message actions as below. Integer arguments preferred; floats (QLab network cues) are accepted.

Command reference

Address	Arguments	Action
`/pts/estop`	-	E-STOP every mount
`/pts/cam/N/estop`	-	E-STOP mount N
`/pts/cam/N/goto`	`slot 1-10`	Recall a stored position (uses the mount's active speed presets)
`/pts/cam/N/store`	`slot 1-10`	Store the current position
`/pts/cam/N/clear`	`slot 1-10`	Clear a stored position
`/pts/cam/N/jog`	`pan tilt slider zoom`	Velocities -1000..1000; re-streamed at 20 Hz until stopped
`/pts/cam/N/jog/stop`	-	Stop jogging (put on the button's release action)
`/pts/cam/N/speed/pt`	`1-4`	Active pan/tilt speed preset
`/pts/cam/N/speed/sl`	`1-4`	Active slider speed preset
`/pts/cam/N/subject`	`0-7`	Select look-at subject; switches live during a move
`/pts/cam/N/lookat`	`0 or 1`	Look-at slider move to min (0) or max (1) with the selected subject

N = camera 1-5. Slots and speed presets are 1-based, subjects 0-based - matching every screen in the system.

Button recipes

Button	Setup
Recall shot 3, cam 2	Press: `/pts/cam/2/goto` with int 3
Hold-to-jog pan-left, cam 1	Press: `/pts/cam/1/jog` ints -400 0 0 0 Release: `/pts/cam/1/jog/stop`
Look-at pair, cam 3	Button A: `/pts/cam/3/subject` int 2 Button B: `/pts/cam/3/lookat` int 1 (press A mid-move to retarget live)
QLab cue	Network cue to port 9700: `/pts/cam/2/goto 4` - camera moves fire from the cue stack, in time with light and sound.
Show-stopper	A big red `/pts/estop` (no argument).

Safety

- OSC jogs are re-streamed at 20 Hz with a 15 s TTL, so a lost release message cannot run a mount away; prefer goto / lookat for long moves.
- The mounts' own 500 ms dead-man is the final backstop: if the hub or PC dies mid-jog, motion stops regardless of the controller.
- E-STOP over OSC clears held jog streams before broadcasting the stop.